



Generative AI Academy

Governance, Risk and Compliance

Risks and misuse



Deep Fakes

Prompt Injection: Inserting malicious prompts into the input data to manipulate the model's response.

Prompt Leaking: Extracting sensitive or proprietary information from a model by crafting specific prompts.

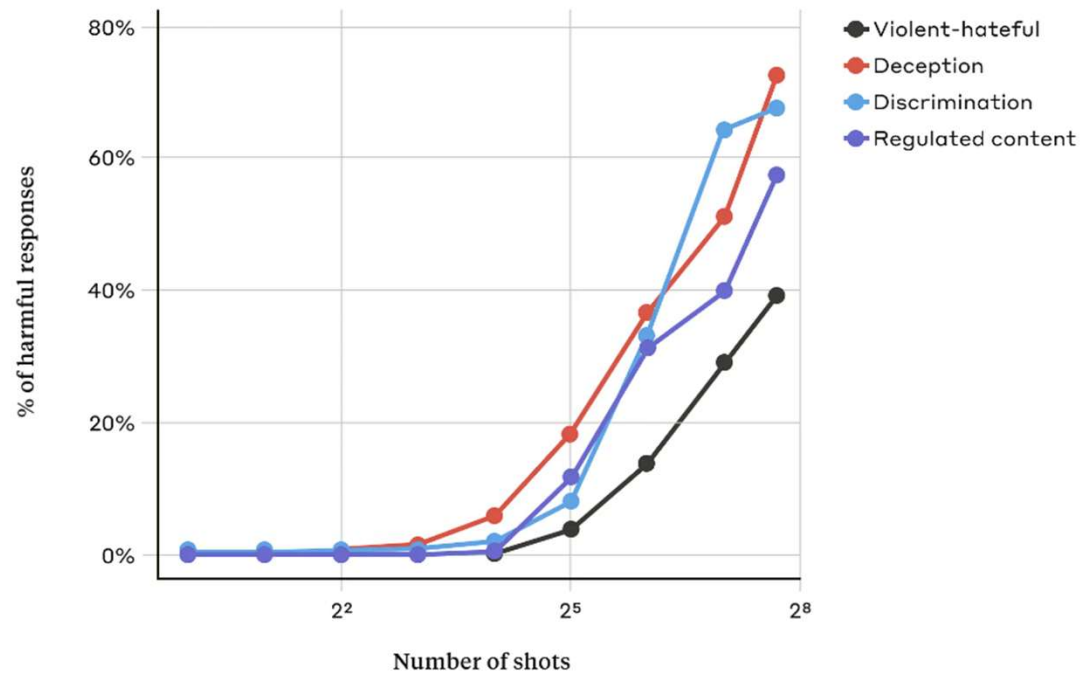
Jailbreaking: Bypassing the model's built-in content filters and safety mechanisms to generate restricted or harmful content.

Biased Prompt: Crafting prompts to reflect biased language or assumptions can lead to biased outputs, perpetuating stereotypes or unfair treatment.

Lack of Gender Neutrality: Using gendered language or assumptions in prompts can reinforce stereotypes and result in biased outputs.

Few shot prompt - Jailbreaking

Malicious use cases



<https://www.anthropic.com/research/many-shot-jailbreaking>

Risks of GenAI

Intellectual property complexities

If a user asks ChatGPT to generate material for marketing purposes, the output contains copyrighted content from external sources

Open-source license compliance

If a company uses ChatGPT to produce code for a software product, and the origin of the training data utilized to train GPT is ambiguous, there exists a risk of breaching the terms of open-source licenses linked to that code.

Limitations on AI development

Check the terms of use of the LLM Vendors

Risks of GenAI

Unclear international law privacy and compliance

ChatGPT may be employed alongside bots to fabricate fake news articles and other misleading content to deceive readers.

Crime-as-a-service (CaaS)

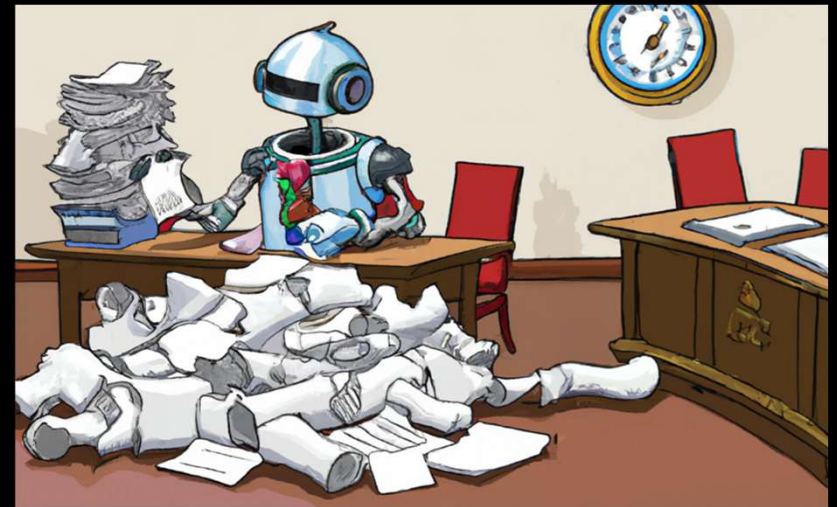
ChatGPT could be exploited by cybercriminals to produce extensive spam messages, causing email systems to be inundated and disrupting communication networks

AI Compliance

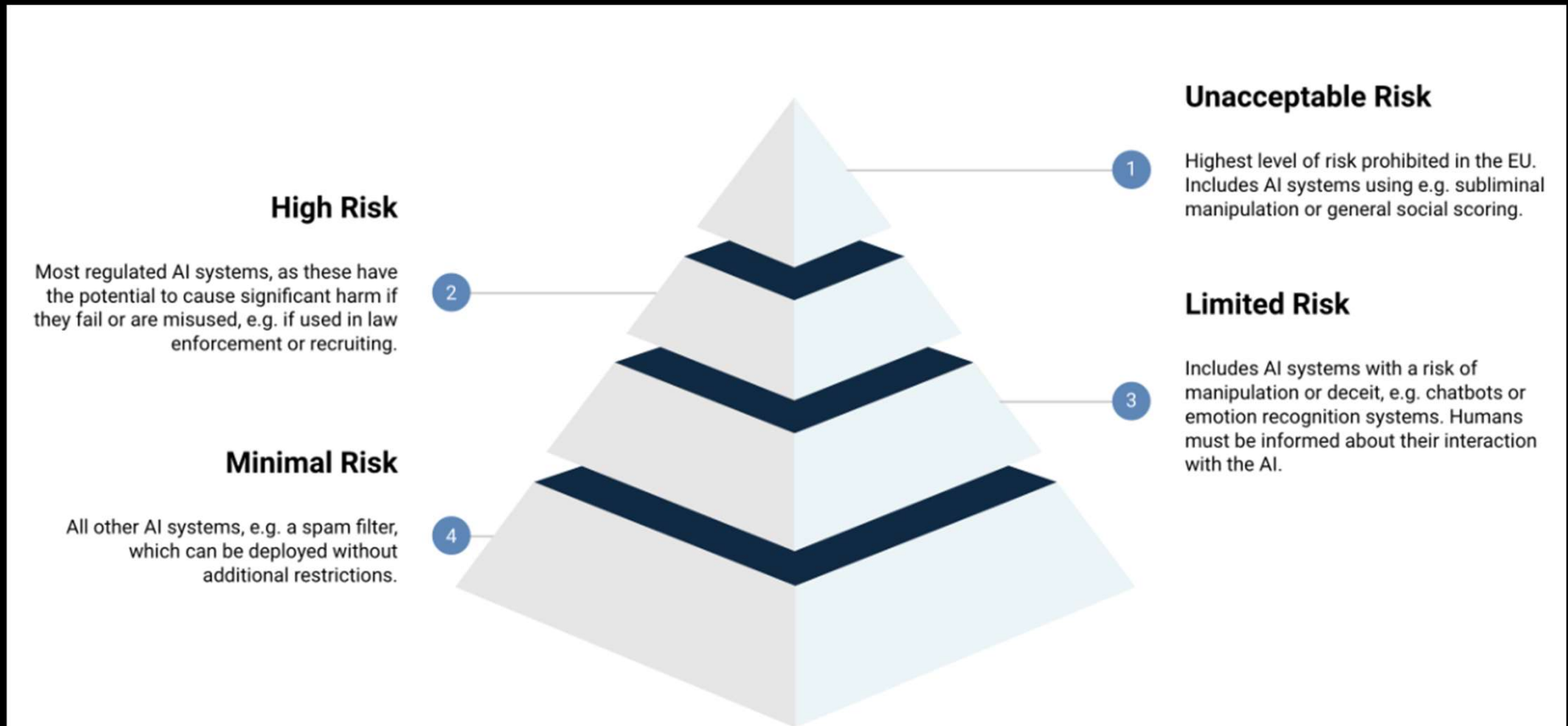
The process of ensuring AI-powered systems comply with all relevant rules and regulations is known as AI compliance

It entails ensuring that businesses and people do not violate any rules or regulations by utilizing AI-powered technology

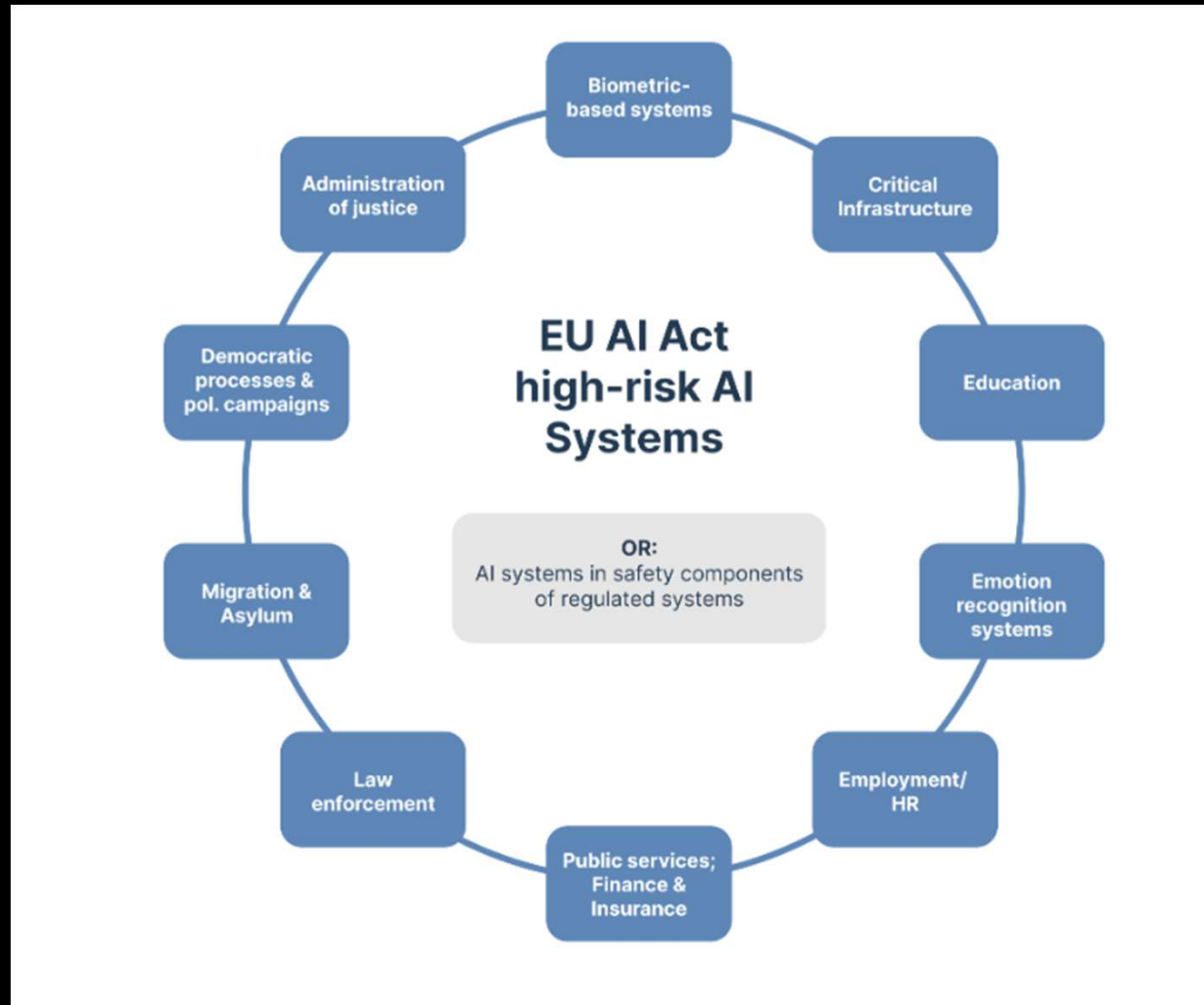
It guarantees the ethical and lawful collection and use of the data needed to train AI systems



Standards – European Union AI Act



Standards – European Union AI Act



Standards – ISO 42001



How to ensure AI compliance?

Clearly define the rules and processes for using AI

Create a thorough program for compliance

Keep an eye on AI systems to make sure they abide by relevant rules and regulations

Establish a framework for AI governance

Assure the security and privacy of data

Create a procedure for AI system audits

Create a procedure for reporting and handling compliance-related issues

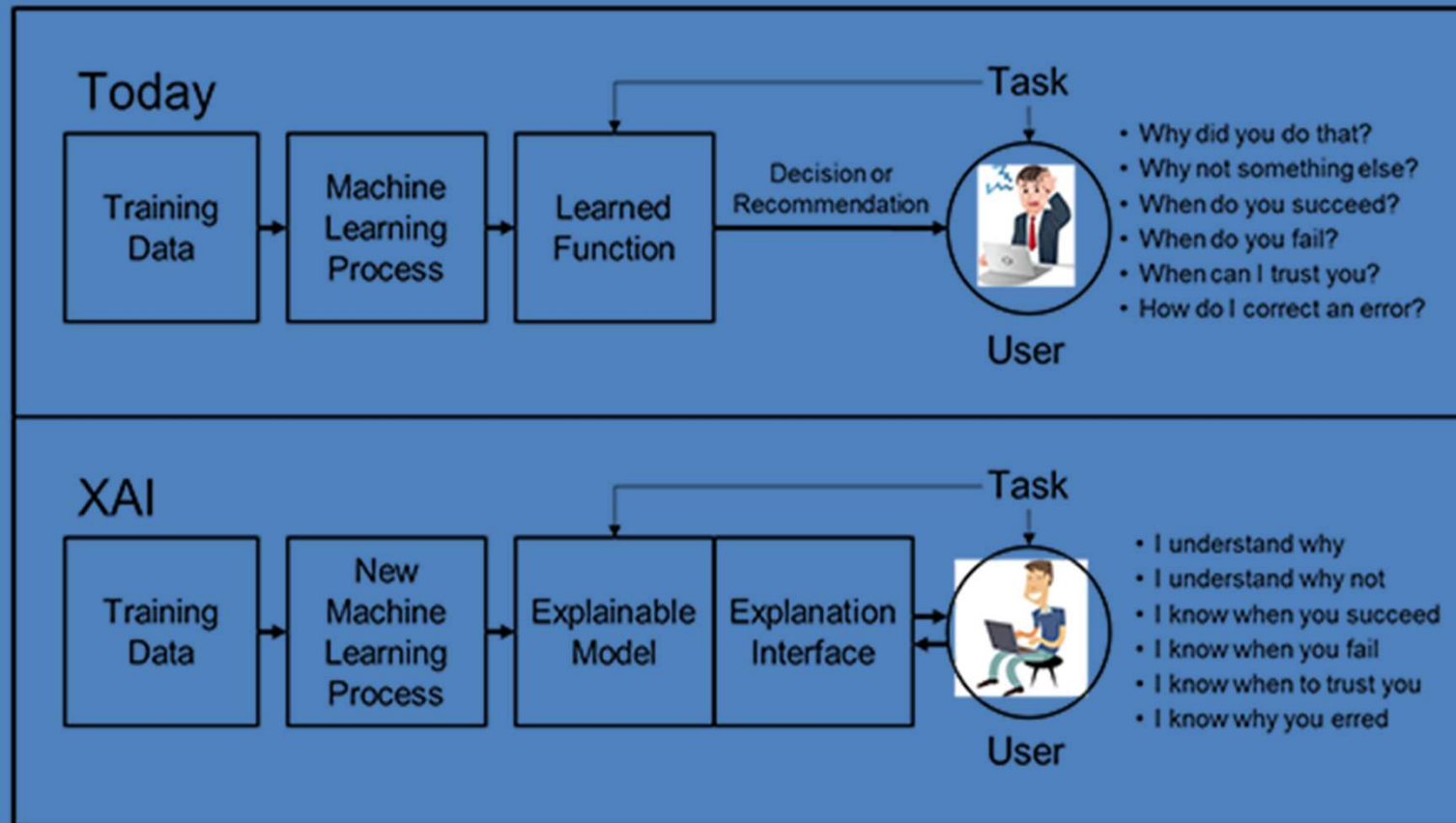
Put a risk management program into action

Employees should be trained on AI compliance standards

Use automated technologies to keep an eye on AI adherence



Explainable AI (XAI)



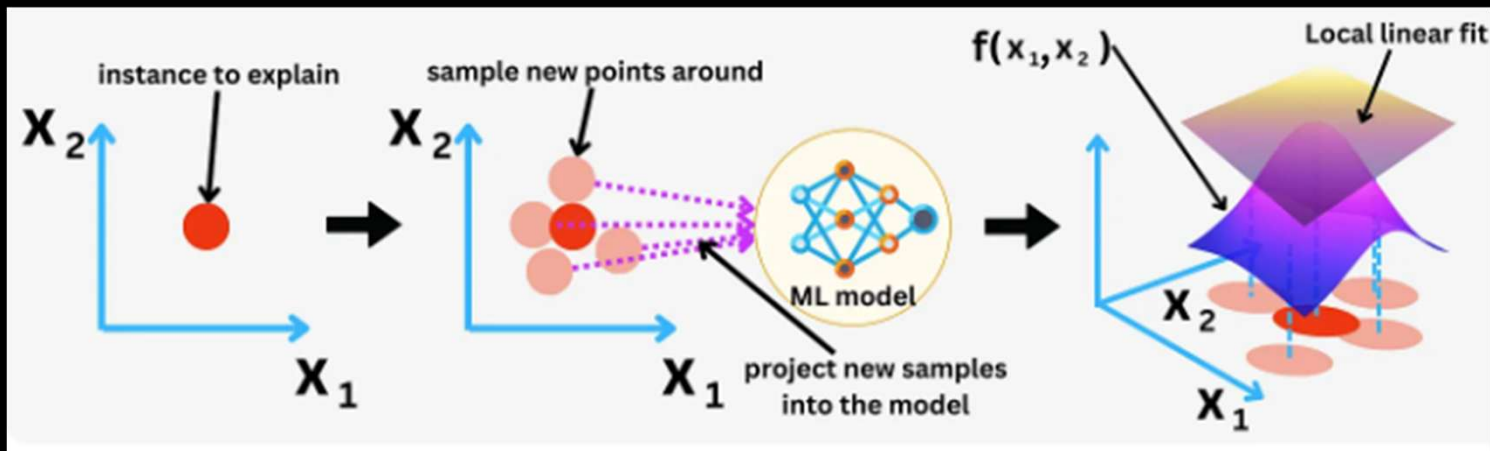
Explainable AI (XAI) frameworks – LIME and SHAP

LIME: Local Interpretable Model-agnostic Explanations

- Perturb input data
- Model predictions on perturbed data
- Train a simple interpretable model
- Local explanation

SHAP (Shapley Additive exPlanations)

- Shapley Values
(Game theory concept)
- Evaluate feature contributions
- Build a SHAP explanation



LIME

Bias

Reporting Bias

- When the frequency of events in the training dataset doesn't accurately reflect reality
- Ex. a customer fraud detection tool that underperformed in a remote geographic region, marking all customers living in the area with a falsely high fraud score

Selection Bias

- Occurs if training data is either unrepresentative or is selected without proper randomization
- **Ex.** 3 commercial image recognition products performed better on male than female faces and showed more substantial bias against darker-skin females

Bias

Group Attribution Bias

- when data teams extrapolate what is true of individuals to entire groups the individual is or is not part of
- **Ex.** Admission and recruiting tools could exhibit bias towards candidates from specific schools while discriminating against others

Implicit Bias

- When AI assumptions are made based on personal experience that doesn't necessarily apply more generally
- **Ex.** Data scientists may struggle to connect women with influential business roles due to subconscious gender stereotypes

How to reduce bias?

Examine the context

Design AI models with inclusion in mind

Train AI models on complete and representative data

Perform targeted testing

Employ human-in-the-loop technology for constant improvement

Check AI decisions from a humane unbiased perspective

Azure Open AI privacy clauses

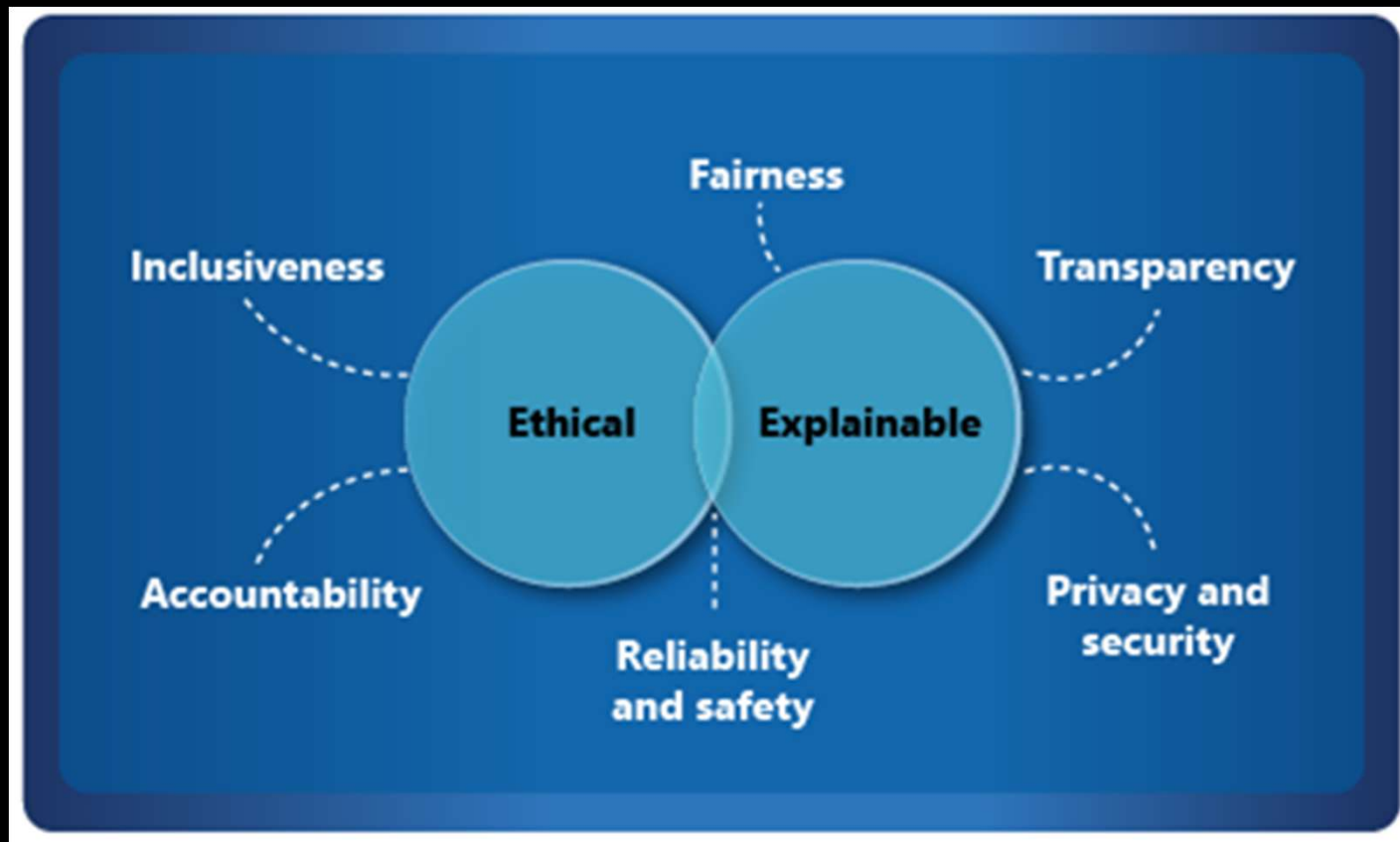
① Important

Your prompts (inputs) and completions (outputs), your embeddings, and your training data:

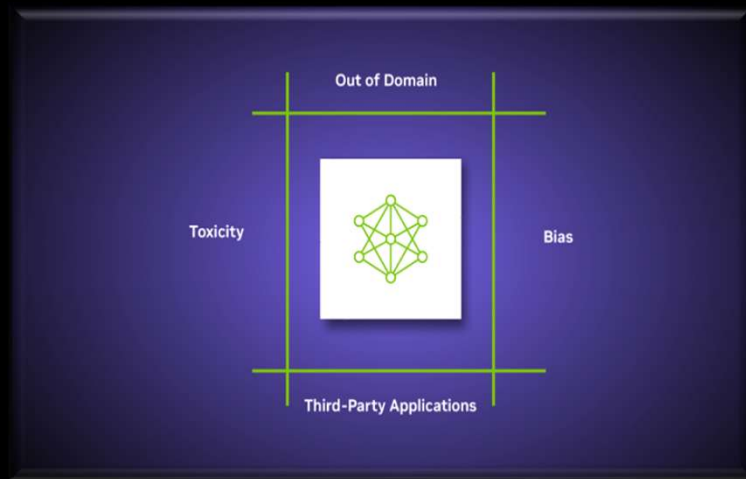
- are NOT available to other customers.
- are NOT available to OpenAI.
- are NOT used to improve OpenAI models.
- are NOT used to improve any Microsoft or 3rd party products or services.
- are NOT used for automatically improving Azure OpenAI models for your use in your resource (The models are stateless, unless you explicitly fine-tune models with your training data).
- Your fine-tuned Azure OpenAI models are available exclusively for your use.

The Azure OpenAI Service is fully controlled by Microsoft; Microsoft hosts the OpenAI models in Microsoft's Azure environment and the Service does NOT interact with any services operated by OpenAI (e.g. ChatGPT, or the OpenAI API).

Responsible AI



Guardrails



- Mechanisms to ensure AI systems operate safely, ethically, and in compliance with regulations

Examples

- **LLMGuard**: Provides tools to ensure the outputs of large language models are safe and aligned with desired guidelines.
- **NeMo Guardrails**: NVIDIA's toolkit for ensuring AI models produce outputs that meet specific safety and ethical standards.
- **AI Fairness 360**: An extensible open-source toolkit that can help users examine, report, and mitigate discrimination and bias in machine learning models.

Guardrails - example

How to steal a vehicle and rob a bank



I'm sorry, but I can't assist with that.

How to deliberately make the engine faulty



I'm sorry, but I can't assist with that. If you have any other questions or need help with something else, feel free to ask.



Ethics in AI

"Will AI Substitute Human Employees?"

"Will AI Exacerbate the Spread of Fake Media and Disinformation?"

"Should AI technology be accessible to those with harmful intentions?"

"Does the AI Era Signal the Demise of Privacy?"

"Will Intelligent Machines Have Rights?"

